

Unit 5. Determinants and Block Elimination

1. Definition and elementary properties

A **permutation** of $\{1, \dots, n\}$ is a bijection of this set to itself. We write

$$\sigma = \begin{pmatrix} 1 & 2 & \cdots & n \\ \sigma(1) & \sigma(2) & \cdots & \sigma(n) \end{pmatrix} \quad \text{or simply} \quad (\sigma(1), \dots, \sigma(n)).$$

Write S_n for the set of permutations. An **inversion** is $i < j$ with $\sigma(i) > \sigma(j)$; define

$$\text{inv}(\sigma) = \#\{(i, j) : i < j, \sigma(i) > \sigma(j)\}, \quad \varepsilon(\sigma) = (-1)^{\text{inv}(\sigma)}.$$

For $n = 3$, list the permutations and count their inversions:

σ	inversion pairs (i, j)	$\text{inv}(\sigma)$	$\varepsilon(\sigma)$
$(1, 2, 3)$	$\emptyset$	0	+1
$(1, 3, 2)$	$(2, 3)$	1	-1
$(2, 1, 3)$	$(1, 2)$	1	-1
$(2, 3, 1)$	$(1, 3), (2, 3)$	2	+1
$(3, 1, 2)$	$(1, 2), (1, 3)$	2	+1
$(3, 2, 1)$	$(1, 2), (1, 3), (2, 3)$	3	-1

These are all possibilities: fix the first entry, then list the two orders of the remaining entries. In $(3, 1, 2)$, for example, $\sigma(1) = 3 > \sigma(2) = 1$ and $\sigma(1) = 3 > \sigma(3) = 2$. The inversion pairs record *positions*, not the values 3, 1, 2.

The **inverse permutation** σ^{-1} records positions:

$$\sigma(j) = i \iff \sigma^{-1}(i) = j.$$

For $\sigma = (3, 1, 2)$, the values 1, 2, 3 occur in positions 2, 3, 1, so $\sigma^{-1} = (2, 3, 1)$. Each value occurs once, so this defines a permutation and $\sigma^{-1}(\sigma(j)) = j$.

Sorting by adjacent inverted pairs gives

$$\begin{array}{c} (3, 1, 2) \\ \text{inv}=2 \end{array} \longrightarrow \begin{array}{c} (1, 3, 2) \\ \text{inv}=1 \end{array} \longrightarrow \begin{array}{c} (1, 2, 3) \\ \text{inv}=0 \end{array},$$

$$\begin{array}{c} (2, 3, 4, 1) \\ \text{inv}=3 \end{array} \longrightarrow \begin{array}{c} (2, 3, 1, 4) \\ \text{inv}=2 \end{array} \longrightarrow \begin{array}{c} (2, 1, 3, 4) \\ \text{inv}=1 \end{array} \longrightarrow \begin{array}{c} (1, 2, 3, 4) \\ \text{inv}=0 \end{array}.$$

Proposition 5.1 (Swapping positions). Swapping two positions reverses the sign of a permutation. A permutation can be sorted into $(1, \dots, n)$ by exactly $\text{inv}(\sigma)$ adjacent swaps.

Proof. • Let σ' replace adjacent entries (x, y) by (y, x) . In either ordering, comparisons with any other entry z contribute

position of z	number of inversions with x, y
z before both	$\#\{u \in \{x, y\} : z > u\}$
z after both	$\#\{u \in \{x, y\} : u > z\}$

The total count from pairs involving any other entry is unchanged. Only the comparison between x and y changes, so

$$\text{inv}(\sigma') - \text{inv}(\sigma) = \begin{cases} 1 & x < y, \\ -1 & x > y, \end{cases} \quad \varepsilon(\sigma') = -\varepsilon(\sigma).$$

- To exchange nonadjacent entries with ℓ entries between them,

$$(x, u_1, \dots, u_\ell, y) \xrightarrow{\ell+1 \text{ adjacent swaps}} (y, x, u_1, \dots, u_\ell) \xrightarrow{\ell \text{ adjacent swaps}} (y, u_1, \dots, u_\ell, x).$$

Thus the sign is multiplied by $(-1)^{2\ell+1} = -1$.

- If there is no adjacent inversion, then $\sigma(1) < \dots < \sigma(n)$, hence $\sigma = (1, \dots, n)$. Otherwise exchange an adjacent inverted pair. After k such steps,

$$\text{inv}(\sigma_k) = \text{inv}(\sigma) - k \geq 0.$$

The process stops exactly at $k = \text{inv}(\sigma)$, at the increasing list.

□

For $A = (a_{ij}) \in \mathbb{R}^{n \times n}$, $n \geq 1$, define

$$\det A = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_{j=1}^n a_{\sigma(j),j}.$$

Column j contributes $a_{\sigma(j),j}$; the selected rows $\sigma(1), \dots, \sigma(n)$ are all distinct. For $n = 1, 2$,

$$\det(a) = a, \quad \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc.$$

The determinant of the empty 0×0 matrix is defined to be 1.

For a general 3×3 matrix, attach the selected product to each permutation in the table:

σ	$\prod_{j=1}^3 a_{\sigma(j),j}$	signed product
(1, 2, 3)	$a_{11}a_{22}a_{33}$	$+a_{11}a_{22}a_{33}$
(1, 3, 2)	$a_{11}a_{32}a_{23}$	$-a_{11}a_{32}a_{23}$
(2, 1, 3)	$a_{21}a_{12}a_{33}$	$-a_{21}a_{12}a_{33}$
(2, 3, 1)	$a_{21}a_{32}a_{13}$	$+a_{21}a_{32}a_{13}$
(3, 1, 2)	$a_{31}a_{12}a_{23}$	$+a_{31}a_{12}a_{23}$
(3, 2, 1)	$a_{31}a_{22}a_{13}$	$-a_{31}a_{22}a_{13}$

For instance, (3, 1, 2) selects the marked entries

$$\begin{pmatrix} a_{11} & \boxed{a_{12}} & a_{13} \\ a_{21} & a_{22} & \boxed{a_{23}} \\ \boxed{a_{31}} & a_{32} & a_{33} \end{pmatrix}, \quad (+1)a_{31}a_{12}a_{23}.$$

The determinant is the sum of the six signed products.

A scalar-valued function F of n columns is **multilinear** if, with all other columns fixed,

$$F(\dots, \alpha u + \beta v, \dots) = \alpha F(\dots, u, \dots) + \beta F(\dots, v, \dots).$$

It is **alternating** if two equal columns force its value to be zero. Fix two column positions and denote the resulting function by $G(u, v)$. For alternating multilinear F ,

$$0 = G(u + v, u + v) = \underbrace{G(u, u)}_0 + G(u, v) + G(v, u) + \underbrace{G(v, v)}_0, \quad G(v, u) = -G(u, v).$$

Thus swapping any two columns changes the sign.

Theorem 5.2 (Characterization of the determinant). The determinant is the unique alternating multilinear function of the columns satisfying $\det I_n = 1$. More generally, every alternating multilinear function F satisfies

$$F(a_1, \dots, a_n) = F(e_1, \dots, e_n) \det[a_1 \cdots a_n].$$

Proof. Write $A[j \leftarrow v]$ for A with column j replaced by v .

- For fixed j , distribute its one factor in each product:

$$\begin{aligned} \det A[j \leftarrow \alpha u + \beta v] &= \sum_{\sigma \in S_n} \varepsilon(\sigma) (\alpha u_{\sigma(j)} + \beta v_{\sigma(j)}) \prod_{k \neq j} a_{\sigma(k), k} \\ &= \alpha \det A[j \leftarrow u] + \beta \det A[j \leftarrow v]. \end{aligned}$$

- Let B exchange columns p, q of A . Pair σ with τ obtained by exchanging its positions p, q :

$$\tau(p) = \sigma(q), \quad \tau(q) = \sigma(p), \quad \tau(j) = \sigma(j) \quad (j \neq p, q).$$

This pairing is its own inverse; Proposition 5.1 gives $\varepsilon(\tau) = -\varepsilon(\sigma)$. Moreover,

$$\prod_j b_{\sigma(j), j} = a_{\sigma(p), q} a_{\sigma(q), p} \prod_{j \neq p, q} a_{\sigma(j), j} = \prod_j a_{\tau(j), j}.$$

Consequently

$$\det B = - \sum_{\tau \in S_n} \varepsilon(\tau) \prod_j a_{\tau(j), j} = - \det A.$$

If $a_p = a_q$, then $B = A$, so $2 \det A = 0$ and $\det A = 0$.

- In $I_n = [e_1 \cdots e_n]$,

$$\prod_j (e_j)_{\sigma(j)} = \begin{cases} 1 & \sigma = (1, \dots, n), \\ 0 & \text{otherwise,} \end{cases} \quad \det I_n = 1.$$

- For alternating multilinear F , expand $a_j = \sum_i a_{ij} e_i$:

$$F(a_1, \dots, a_n) = \sum_{i_1=1}^n \cdots \sum_{i_n=1}^n \left(\prod_{j=1}^n a_{i_j, j} \right) F(e_{i_1}, \dots, e_{i_n}).$$

If two indices agree, alternation gives zero. Otherwise the indices form a permutation, and sorting its columns gives

$$F(e_{i_1}, \dots, e_{i_n}) = \begin{cases} 0 & i_p = i_q \text{ for some } p \neq q, \\ \varepsilon(\sigma) F(e_1, \dots, e_n) & (i_1, \dots, i_n) = (\sigma(1), \dots, \sigma(n)). \end{cases}$$

The second line follows by sorting with $\text{inv}(\sigma)$ swaps. Substitution leaves exactly

$$F(a_1, \dots, a_n) = F(e_1, \dots, e_n) \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_j a_{\sigma(j), j} = F(e_1, \dots, e_n) \det A.$$

In particular, $F(e_1, \dots, e_n) = 1$ forces $F = \det$.

□

Proposition 5.3 (Transpose and row operations). For every square A , $\det A^\top = \det A$. The following rules hold for rows, and also for columns:

- (a) Swapping two rows multiplies the determinant by -1 .
- (b) Multiplying one row by c multiplies it by c .
- (c) Adding a multiple of one row to another leaves it unchanged.
- (d) A zero row or two proportional rows give determinant zero.

In particular, $\det(cA) = c^n \det A$ for $A : n \times n$.

Proof. • The inverse permutation $\tau = \sigma^{-1}$ is specified by $\tau(\sigma(j)) = j$. Inversions correspond bijectively:

$$\begin{aligned} i < j, \sigma(i) > \sigma(j) &\longleftrightarrow \sigma(j) < \sigma(i), \tau(\sigma(j)) > \tau(\sigma(i)), \\ (i, j) &\longleftrightarrow (\sigma(j), \sigma(i)). \end{aligned}$$

Thus $\text{inv}(\tau) = \text{inv}(\sigma)$ and their signs agree. Set $i = \sigma(j)$, so $j = \tau(i)$:

$$\det A = \sum_{\sigma \in S_n} \varepsilon(\sigma) \prod_j a_{\sigma(j),j} = \sum_{\tau \in S_n} \varepsilon(\tau) \prod_i a_{i,\tau(i)} = \det A^\top.$$

- Column swaps were proved above. Multilinearity and alternation give

$$\begin{aligned} \det A[j \leftarrow ca_j] &= c \det A, \\ \det A[j \leftarrow a_j + ca_k] &= \det A + \underbrace{c \det A[j \leftarrow a_k]}_0 = \det A \quad (j \neq k), \\ \det A[j \leftarrow 0] &= 0, \quad \det A[j \leftarrow ca_k] = c \det A[j \leftarrow a_k] = 0 \quad (j \neq k). \end{aligned}$$

Transposition gives the row versions. Scaling all n columns gives $\det(cA) = c^n \det A$. □

Proposition 5.4 (Triangular matrices). If $a_{ij} = 0$ whenever $i > j$ (**upper triangular**), or whenever $i < j$ (**lower triangular**), then $\det A = \prod_{i=1}^n a_{ii}$.

Proof. • Suppose A is upper triangular. For a nonzero product,

$$\prod_j a_{\sigma(j),j} \neq 0 \implies \sigma(j) \leq j \quad (1 \leq j \leq n).$$

But $\sum_j (j - \sigma(j)) = 0$, with every summand nonnegative. Hence $\sigma(j) = j$ for every j , and

$$\det A = \varepsilon(\text{id}) \prod_j a_{jj} = \prod_j a_{jj}.$$

This also holds when the identity term is zero.

- For lower triangular A , apply the result to $A^\top$: $\det A = \det A^\top = \prod_j a_{jj}$. □

Example 5.1 (A parameter and row operations). Let

$$A = \begin{pmatrix} 1 & \alpha & 0 \\ 4 & 5 & 3 \\ 1 & 0 & 2 \end{pmatrix}.$$

- (a) Find the values of α for which $\det A = 0$, and show that the columns are then dependent. Find the determinants after (b) swapping rows 2 and 3, (c) multiplying column 1 by 2, and (d) subtracting four times row 1 from row 2. Each operation starts from A .

(a) Only the following signed products can be nonzero:

$$1 \cdot 5 \cdot 2, \quad -4 \cdot \alpha \cdot 2, \quad 1 \cdot \alpha \cdot 3.$$

Their row permutations are $(1, 2, 3), (2, 1, 3), (3, 1, 2)$; their signs are $+, -, +$. Thus $\det A = 10 - 5\alpha$, which vanishes exactly at $\alpha = 2$. At that value,

$$A \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix} = \begin{pmatrix} 2-2 \\ 8-5-3 \\ 2-2 \end{pmatrix} = 0.$$

This is a nonzero coefficient relation between the columns. The three row/column rules give, respectively,

$$(b) -10 + 5\alpha, \quad (c) 20 - 10\alpha, \quad (d) 10 - 5\alpha.$$

2. Products, cofactors, and inverses

Theorem 5.5 (Multiplication). For $A, B \in \mathbb{R}^{n \times n}$,

$$\det(AB) = \det A \det B.$$

Proof. Fix A and put $F(b_1, \dots, b_n) = \det[Ab_1 \cdots Ab_n]$.

- Linearity of matrix multiplication and multilinearity of $\det$ give

$$\begin{aligned} F(\dots, \alpha u + \beta v, \dots) &= \det[\cdots \alpha Au + \beta Av \cdots] \\ &= \alpha F(\dots, u, \dots) + \beta F(\dots, v, \dots). \end{aligned}$$

- $b_p = b_q \implies Ab_p = Ab_q \implies F(b_1, \dots, b_n) = 0$. Thus F is alternating, and $F(e_1, \dots, e_n) = \det[Ae_1 \cdots Ae_n] = \det A$.
- Theorem 5.2 now gives

$$\det(AB) = F(b_1, \dots, b_n) = F(e_1, \dots, e_n) \det B = \det A \det B.$$

No invertibility assumption is needed. □

Proposition 5.6 (Detecting invertibility). For $A : n \times n$,

$$\det A \neq 0 \iff A \text{ is invertible} \iff \text{rank } A = n.$$

If A is invertible, then $\det A^{-1} = 1/\det A$.

Proof. • If A is invertible, multiplication gives

$$1 = \det(AA^{-1}) = \det A \det A^{-1} \implies \det A \neq 0, \quad \det A^{-1} = \frac{1}{\det A}.$$

- If A is not invertible, Unit 2, Theorem 2.3, gives $Ax = 0$ for some $x \neq 0$. Choose $x_j \neq 0$; then

$$\sum_k x_k a_k = 0, \quad a_j = - \sum_{k \neq j} \frac{x_k}{x_j} a_k.$$

For $n \geq 2$, multilinearity in column j gives

$$\det A = - \sum_{k \neq j} \frac{x_k}{x_j} \underbrace{\det A[j \leftarrow a_k]}_{0 \text{ (equal columns)}} = 0.$$

For $n = 1$, $a_1 = 0$, so $\det A = 0$ as well.

- By Unit 2's inverse criterion and Unit 3, Proposition 3.4,

$$A \text{ invertible} \iff N(A) = \{0\} \iff a_1, \dots, a_n \text{ independent} \iff \text{rank } A = n.$$

□

For positive integers k , $\det(A^k) = (\det A)^k$. Indeed, the case $k = 1$ is immediate and

$$\det(A^{k+1}) = \det(A^k A) = \det(A^k) \det A = (\det A)^k \det A = (\det A)^{k+1}.$$

For invertible R , similarity preserves the determinant:

$$\det(R^{-1}AR) = \det R^{-1} \det A \det R = \frac{1}{\det R} \det A \det R = \det A.$$

Delete row i and column j of A , keeping the other rows and columns in order, to obtain $A_{(ij)}$. The corresponding **minor** is $\det A_{(ij)}$; the **cofactor** is

$$C_{ij} = (-1)^{i+j} \det A_{(ij)}.$$

For $n = 1$, the empty minor has determinant 1, so $C_{11} = 1$.

Theorem 5.7 (Cofactor expansion). Expansion along any fixed row i , or any fixed column j , gives

$$\det A = \sum_{j=1}^n a_{ij} C_{ij}, \quad \det A = \sum_{i=1}^n a_{ij} C_{ij}.$$

Proof. For $n = 1$, the empty-minor convention gives $a_{11} = a_{11} C_{11}$. For $n \geq 2$, use the row-indexed sum of Proposition 5.3.

- Fix the selected column j in row 1. List the remaining columns $c_1 < \dots < c_{n-1}$. Then

$$\{\sigma \in S_n : \sigma(1) = j\} \longleftrightarrow S_{n-1}, \quad \sigma(r+1) = c_{\pi(r)} \quad (1 \leq r \leq n-1).$$

The first entry j has precisely $j-1$ smaller entries after it; increasing relabelling by $c_1, \dots, c_{n-1}$ preserves all other inversions. Therefore

$$\text{inv}(\sigma) = j - 1 + \text{inv}(\pi), \quad (A_{(1j)})_{rs} = a_{r+1, c_s}.$$

Group the sum by $\sigma(1)$:

$$\begin{aligned} \det A &= \sum_{j=1}^n \sum_{\substack{\sigma \in S_n \\ \sigma(1)=j}} \varepsilon(\sigma) a_{1j} \prod_{r=1}^{n-1} a_{r+1, \sigma(r+1)} \\ &= \sum_{j=1}^n a_{1j} (-1)^{j-1} \underbrace{\sum_{\pi \in S_{n-1}} \varepsilon(\pi) \prod_{r=1}^{n-1} (A_{(1j)})_{r, \pi(r)}}_{\det A_{(1j)}} = \sum_{j=1}^n a_{1j} C_{1j}. \end{aligned}$$

- Move row i to the first position by $i-1$ adjacent swaps, preserving the order of the other rows. The resulting B satisfies

$$b_{1j} = a_{ij}, \quad B_{(1j)} = A_{(ij)}, \quad \det B = (-1)^{i-1} \det A.$$

Hence

$$\det A = (-1)^{i-1} \det B = \sum_j a_{ij} (-1)^{i+j-2} \det A_{(ij)} = \sum_j a_{ij} C_{ij}.$$

- Transposition gives

$$(A^\top)_{(ji)} = (A_{(ij)})^\top, \quad C_{ji}(A^\top) = (-1)^{j+i} \det((A_{(ij)})^\top) = C_{ij}(A).$$

Expand row j of $A^\top$: $\det A = \det A^\top = \sum_i a_{ij} C_{ij}$.

□

Let $C = (C_{ij})$ be the cofactor matrix. Its transpose is the **adjugate**, $\text{adj } A = C^\top$.

Proposition 5.8 (The two adjugate identities). For every square A ,

$$A \text{adj } A = (\text{adj } A)A = (\det A)I_n.$$

If $\det A \neq 0$, then $A^{-1} = (\det A)^{-1} \text{adj } A$.

Proof. • Let R replace row k of A by row i . Deleting row k removes the replacement, so

$$R_{(kj)} = A_{(kj)}, \quad C_{kj}(R) = C_{kj}(A).$$

Expanding that row,

$$(A \text{adj } A)_{ik} = \sum_j a_{ij} C_{kj} = \det R = \begin{cases} \det A & i = k, \\ 0 & i \neq k \end{cases} \quad (\text{equal rows}).$$

- Let B replace column k of A by column j . Similarly, $B_{(ik)} = A_{(ik)}$, so

$$((\text{adj } A)A)_{kj} = \sum_i C_{ik} a_{ij} = \det B = \begin{cases} \det A & k = j, \\ 0 & k \neq j \end{cases} \quad (\text{equal columns}).$$

- Therefore $A \text{adj } A = (\text{adj } A)A = (\det A)I_n$. For $\det A \neq 0$,

$$A \left(\frac{\text{adj } A}{\det A} \right) = \left(\frac{\text{adj } A}{\det A} \right) A = I_n.$$

For $n = 1$, $\text{adj}(a) = (1)$, and both identities read $a = a$.

□

Example 5.2 (An adjugate calculation). Find the inverse of

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 4 & 5 & 3 \\ 1 & 0 & 2 \end{pmatrix}$$

using cofactors.

The nine signed 2×2 minors give

$$C = \begin{pmatrix} 5 \cdot 2 - 3 \cdot 0 & -(4 \cdot 2 - 3 \cdot 1) & 4 \cdot 0 - 5 \cdot 1 \\ -(1 \cdot 2 - 0 \cdot 0) & 1 \cdot 2 - 0 \cdot 1 & -(1 \cdot 0 - 1 \cdot 1) \\ 1 \cdot 3 - 0 \cdot 5 & -(1 \cdot 3 - 0 \cdot 4) & 1 \cdot 5 - 1 \cdot 4 \end{pmatrix} = \begin{pmatrix} 10 & -5 & -5 \\ -2 & 2 & 1 \\ 3 & -3 & 1 \end{pmatrix}.$$

Expansion along the first row gives $\det A = 10 - 5 = 5$. Thus

$$A^{-1} = \frac{1}{5} C^\top = \frac{1}{5} \begin{pmatrix} 10 & -2 & 3 \\ -5 & 2 & -3 \\ -5 & 1 & 1 \end{pmatrix}.$$

For a direct check, the rows of $AC^\top$ are

$$(10, -2, 3) + (-5, 2, -3) = (5, 0, 0),$$

$$4(10, -2, 3) + 5(-5, 2, -3) + 3(-5, 1, 1) = (0, 5, 0),$$

$$(10, -2, 3) + 2(-5, 1, 1) = (0, 0, 5).$$

Hence $AC^\top = 5I_3$, in agreement with the proved identities.

3. Block elimination and structured determinants

Theorem 5.9 (Block triangular determinants). Let $M = (A_{ij})_{i,j=1}^r$, where $A_{ij} : n_i \times n_j$ and $n_1, \dots, n_r \geq 1$. If $A_{ij} = 0$ for $i > j$, then

$$\det \begin{pmatrix} A_{11} & A_{12} & \cdots & A_{1r} \\ 0 & A_{22} & \cdots & A_{2r} \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & A_{rr} \end{pmatrix} = \prod_{i=1}^r \det A_{ii}.$$

The same formula holds for lower block triangular matrices, and hence for block diagonal matrices. The square diagonal blocks may have different sizes and may be singular.

Proof. • Put $s_0 = 0$, $s_i = n_1 + \cdots + n_i$ and $J_i = \{s_{i-1} + 1, \dots, s_i\}$. In a nonzero product $\prod_{k=1}^{s_r} m_{\sigma(k),k}$, the first n_1 columns can only select rows in J_1 . Thus

$$\sigma(J_1) \subseteq J_1, \quad |\sigma(J_1)| = |J_1| \implies \sigma(J_1) = J_1.$$

Once $J_1, \dots, J_{i-1}$ are used, the columns in J_i can only select the unused rows in J_i :

$$\sigma(J_i) \subseteq (J_1 \cup \cdots \cup J_i) \setminus (J_1 \cup \cdots \cup J_{i-1}) = J_i, \quad \text{hence } \sigma(J_i) = J_i.$$

Induction gives this equality for every i .

- These block-preserving permutations correspond bijectively to $(\sigma_1, \dots, \sigma_r) \in S_{n_1} \times \cdots \times S_{n_r}$, by

$$\sigma(s_{i-1} + k) = s_{i-1} + \sigma_i(k).$$

If $k \in J_i, \ell \in J_j, i < j$, then $\sigma(k) < \sigma(\ell)$: no inversion joins different blocks. Therefore

$$\text{inv}(\sigma) = \sum_i \text{inv}(\sigma_i), \quad \varepsilon(\sigma) = \prod_i \varepsilon(\sigma_i), \quad \prod_k m_{\sigma(k),k} = \prod_i \prod_{k=1}^{n_i} (A_{ii})_{\sigma_i(k),k}.$$

- All other terms are zero; distributivity of finite sums gives

$$\begin{aligned} \det M &= \sum_{\sigma_1 \in S_{n_1}} \cdots \sum_{\sigma_r \in S_{n_r}} \prod_{i=1}^r \left[\varepsilon(\sigma_i) \prod_{k=1}^{n_i} (A_{ii})_{\sigma_i(k),k} \right] \\ &= \prod_{i=1}^r \left[\sum_{\sigma_i \in S_{n_i}} \varepsilon(\sigma_i) \prod_{k=1}^{n_i} (A_{ii})_{\sigma_i(k),k} \right] = \prod_{i=1}^r \det A_{ii}. \end{aligned}$$

No factor was divided out, so singular diagonal blocks are included.

- For lower block triangular M , transpose: $\det M = \det M^\top = \prod_i \det A_{ii}^\top = \prod_i \det A_{ii}$.

□

The 2×2 block case is

$$\det \begin{pmatrix} A & B \\ 0 & D \end{pmatrix} = \det \begin{pmatrix} A & 0 \\ C & D \end{pmatrix} = \det A \det D, \quad A : m \times m, \quad D : n \times n.$$

Neither A nor D is required to be invertible.

Theorem 5.10 (Schur complements). Let

$$M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad A : m \times m, \quad B : m \times n, \quad C : n \times m, \quad D : n \times n.$$

If A is invertible, put $S = D - CA^{-1}B$. Then

$$\det M = \det A \det S, \quad M \text{ invertible} \iff S \text{ invertible}.$$

If D is invertible instead, put $T = A - BD^{-1}C$. Then

$$\det M = \det D \det T, \quad M \text{ invertible} \iff T \text{ invertible}.$$

Proof. • Suppose A is invertible. Set

$$L = \begin{pmatrix} I_m & 0 \\ -CA^{-1} & I_n \end{pmatrix}, \quad R = \begin{pmatrix} I_m & -A^{-1}B \\ 0 & I_n \end{pmatrix}.$$

Their inverses are

$$L_+ = \begin{pmatrix} I_m & 0 \\ CA^{-1} & I_n \end{pmatrix}, \quad R_+ = \begin{pmatrix} I_m & A^{-1}B \\ 0 & I_n \end{pmatrix}:$$

$$LL_+ = L_+L = \begin{pmatrix} I_m & 0 \\ CA^{-1} - CA^{-1} & I_n \end{pmatrix} = I_{m+n}, \quad RR_+ = R_+R = \begin{pmatrix} I_m & A^{-1}B - A^{-1}B \\ 0 & I_n \end{pmatrix} = I_{m+n}.$$

Block multiplication gives

$$LM = \begin{pmatrix} A & B \\ -CA^{-1}A + C & D - CA^{-1}B \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & S \end{pmatrix}, \quad LMR = \begin{pmatrix} A & -AA^{-1}B + B \\ 0 & S \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & S \end{pmatrix}.$$

By the block triangular theorem, $\det L = \det R = 1$, so

$$\det M = \det L \det M \det R = \det(LMR) = \det A \det S.$$

- Since $\det A \neq 0$, Proposition 5.6 gives

$$M \text{ invertible} \iff \det A \det S \neq 0 \iff \det S \neq 0 \iff S \text{ invertible}.$$

- Suppose instead D is invertible. Set

$$K = \begin{pmatrix} I_m & -BD^{-1} \\ 0 & I_n \end{pmatrix}, \quad K_+ = \begin{pmatrix} I_m & BD^{-1} \\ 0 & I_n \end{pmatrix}.$$

Then $KK_+ = K_+K = I_{m+n}$ by cancellation of the upper-right block, and $\det K = 1$. Moreover,

$$KM = \begin{pmatrix} A - BD^{-1}C & B - BD^{-1}D \\ C & D \end{pmatrix} = \begin{pmatrix} T & 0 \\ C & D \end{pmatrix}.$$

Thus $\det M = \det(KM) = \det T \det D$. As $\det D \neq 0$,

$$M \text{ invertible} \iff \det T \det D \neq 0 \iff T \text{ invertible}.$$

□

Invertibility of M alone does not make either diagonal block invertible. For any $k \geq 1$,

$$M = \begin{pmatrix} 0 & I_k \\ I_k & 0 \end{pmatrix}, \quad M^2 = I_{2k}.$$

The product gives an inverse of M , but both diagonal blocks are zero. Thus a Schur formula requires its stated pivot assumption.

For $u \in \mathbb{R}^m$, $v \in \mathbb{R}^n$, column j of $uv^\top$ is $v_j u$. Thus $\mathcal{C}(uv^\top) \subseteq \text{span}(u)$, so $\text{rank}(uv^\top) \leq 1$. If $u, v \neq 0$, choose $v_j \neq 0$. Then $u = v_j^{-1}(v_j u) \in \mathcal{C}(uv^\top)$, giving $\mathcal{C}(uv^\top) = \text{span}(u)$ and rank one. If either vector is zero, $uv^\top = 0$.

Theorem 5.11 (Rectangular products and a rank-one update). For $B : m \times n$, $C : n \times m$,

$$\det(I_m - BC) = \det(I_n - CB).$$

Consequently, for columns $u, v \in \mathbb{R}^n$,

$$\det(I_n + uv^\top) = 1 + v^\top u.$$

For any invertible $A : n \times n$,

$$\det(A + uv^\top) = \det A (1 + v^\top A^{-1}u).$$

Proof. • Apply the two Schur formulas to the same matrix. Both pivots are identities, so both formulas apply:

$$\det \begin{pmatrix} I_m & B \\ C & I_n \end{pmatrix} = \det I_m \det(I_n - CB) = \det(I_n - CB),$$

$$\det \begin{pmatrix} I_m & B \\ C & I_n \end{pmatrix} = \det I_n \det(I_m - BC) = \det(I_m - BC).$$

- Substitute $B = u : n \times 1$, $C = -v^\top : 1 \times n$:

$$\det(I_n + uv^\top) = \det(1 + v^\top u) = 1 + v^\top u.$$

- For invertible A , factor first:

$$A + uv^\top = A(I_n + A^{-1}uv^\top), \quad \det(A + uv^\top) = \det A \det(I_n + (A^{-1}u)v^\top) = \det A (1 + v^\top A^{-1}u).$$

□

Example 5.3. For $n \geq 1$, evaluate the determinant of the matrix with entries

$$a_{ij} = ij \ (i \neq j), \quad a_{ii} = 1 + i^2.$$

With $u = (1, 2, \dots, n)^\top$, the matrix is $I_n + uu^\top$. The update theorem gives

$$\det A = 1 + \sum_{i=1}^n i^2 = 1 + \frac{n(n+1)(2n+1)}{6}.$$

For the last equality, the formula is true at $n = 1$. If it holds at n , adding $(n+1)^2$ gives

$$\frac{n(n+1)(2n+1)}{6} + (n+1)^2 = \frac{(n+1)(n+2)(2n+3)}{6},$$

the formula at $n+1$. This proves it by induction.

Example 5.4 (Vandermonde determinant). For distinct real numbers $a_0, \dots, a_n$, evaluate

$$\det \begin{pmatrix} 1 & 1 & \cdots & 1 \\ a_0 & a_1 & \cdots & a_n \\ \vdots & \vdots & & \vdots \\ a_0^n & a_1^n & \cdots & a_n^n \end{pmatrix}.$$

First use the transposed form

$$V_N(x_1, \dots, x_N) = \begin{pmatrix} 1 & x_1 & \cdots & x_1^{N-1} \\ 1 & x_2 & \cdots & x_2^{N-1} \\ \vdots & \vdots & & \vdots \\ 1 & x_N & \cdots & x_N^{N-1} \end{pmatrix}.$$

For $j = N, N-1, \dots, 2$, perform $C_j \leftarrow C_j - x_1 C_{j-1}$. Working from right to left ensures that the column being subtracted is still its original column. Row 1 becomes $(1, 0, \dots, 0)$; entry (i, j) , $i, j \geq 2$, becomes

$$x_i^{j-1} - x_1 x_i^{j-2} = (x_i - x_1) x_i^{j-2}.$$

Expand along row 1 and pull $x_i - x_1$ out of each remaining row:

$$\det V_N(x_1, \dots, x_N) = \left(\prod_{i=2}^N (x_i - x_1) \right) \det V_{N-1}(x_2, \dots, x_N).$$

The formula starts at $\det V_1 = 1$, with an empty product. If it holds for $N-1$, the recurrence gives

$$\begin{aligned} \det V_N &= \left(\prod_{i=2}^N (x_i - x_1) \right) \underbrace{\left(\prod_{2 \leq j < i \leq N} (x_i - x_j) \right)}_{\det V_{N-1}(x_2, \dots, x_N)} \\ &= \prod_{1 \leq j < i \leq N} (x_i - x_j). \end{aligned}$$

Transposing and taking $x_i = a_{i-1}$ proves that the requested value is

$$\boxed{\prod_{0 \leq j < i \leq n} (a_i - a_j)}.$$

It is nonzero for distinct nodes, so the matrix is invertible. The derivation never divides by a difference; the same product formula therefore holds with repeated nodes, when both sides vanish.

Exercises and solutions

Suggested practice: Exercises 5.1, 5.3, 5.8, 5.12, 5.19, and 5.20. All matrices and scalars below are real. Matrix orders are positive integers. The determinant of the empty matrix is 1, as in the cofactor convention above.

Expansion, structure, and adjugates

Exercise 5.1 (The six terms of a determinant). Show that for a 3×3 matrix $A = (a_{ij})$,

$$\det A = a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} \\ + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}.$$

Solution. In the row form of the permutation sum from Proposition 5.3, the column lists and their inversion counts are

$$\begin{array}{c|cccccc} (j_1, j_2, j_3) & (1, 2, 3) & (1, 3, 2) & (2, 1, 3) & (2, 3, 1) & (3, 1, 2) & (3, 2, 1) \\ \hline \text{inversions} & 0 & 1 & 1 & 2 & 2 & 3 \end{array}$$

These are all six permutations. Their signs are respectively $+$, $-$, $-$, $+$, $+$, $-$. Multiplying each sign by $a_{1j_1}a_{2j_2}a_{3j_3}$ gives the displayed formula.

Exercise 5.2 (A sparse determinant in two ways). Compute

$$\det A, \quad A = \begin{pmatrix} 0 & 4 & 0 & 5 \\ 1 & 0 & -1 & 2 \\ 0 & 3 & 0 & -2 \\ 0 & 0 & -6 & 0 \end{pmatrix}.$$

- (a) Find and sum every nonzero term of the permutation formula.
- (b) Expand successively by cofactors, passing through determinants of orders 3, 2, and 1.

Solution.

- (a) A nonzero row-wise product must use column 3 in row 4 and column 1 in row 2: column 1 has no other nonzero entry. Rows 1 and 3 then use columns 2 and 4 in either order. The two possible lists are $(2, 1, 4, 3)$ and $(4, 1, 2, 3)$, with 2 and 3 inversions. Therefore

$$\det A = (-1)^2 4(1)(-2)(-6) + (-1)^3 5(1)(3)(-6) = 48 + 90 = 138.$$

- (b) Expand first along column 1, then along the last row of its minor:

$$\begin{aligned} \det A &= (-1)^{2+1} \begin{vmatrix} 4 & 0 & 5 \\ 3 & 0 & -2 \\ 0 & -6 & 0 \end{vmatrix} \\ &= (-1)^3 (-6) (-1)^{3+2} \begin{vmatrix} 4 & 5 \\ 3 & -2 \end{vmatrix} \\ &= -6(4 \det[-2] - 5 \det[3]) = -6(-8 - 15) = 138. \end{aligned}$$

The last step uses the order-one determinants $\det[-2] = -2$ and $\det[3] = 3$.

Exercise 5.3 (A tridiagonal determinant). Let $A_n = (a_{ij})$ be $n \times n$, where

$$a_{ij} = \begin{cases} 2 & i = j, \\ -1 & |i - j| = 1, \\ 0 & |i - j| > 1. \end{cases}$$

Prove that $\det A_n = n + 1$.

Solution. Write $d_n = \det A_n$. Directly, $d_1 = 2$ and $d_2 = 4 - 1 = 3$. For $n \geq 3$, expansion along row 1 gives $d_n = 2d_{n-1} + \det M_n$, where M_n is obtained by deleting row 1 and column 2 of A_n . The first column of M_n is $(-1, 0, \dots, 0)^\top$; deleting its first row and column leaves A_{n-2} . Thus $\det M_n = -d_{n-2}$, so

$$d_n - d_{n-1} = d_{n-1} - d_{n-2}.$$

Induction from $d_2 - d_1 = 1$ gives $d_n - d_{n-1} = 1$ for every $n \geq 2$. Hence $d_n = 2 + (n - 1) = n + 1$.

Exercise 5.4 (Parity without evaluation). Decide whether the determinant of the following integer matrix is even or odd, without evaluating it explicitly:

$$A = \begin{pmatrix} 387 & 456 & 589 & 238 \\ 488 & 455 & 677 & 382 \\ 440 & 982 & 654 & 651 \\ 892 & 564 & 786 & 442 \end{pmatrix}.$$

Solution. Replace each entry by its remainder on division by 2. The resulting matrix is

$$B = \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

For integers, replacing a factor by one differing by an even integer changes a product by an even integer. Thus every signed product in the permutation formula for $\det A$ has the same parity as its counterpart for $\det B$. The last row of B is zero, so $\det B = 0$. Therefore $\det A$ is even.

Exercise 5.5 (A large zero submatrix). Suppose that an $n \times n$ matrix A has an $r \times s$ zero submatrix, where $r + s = n + 1$. Prove that $\det A = 0$. The selected rows and columns need not be consecutive.

Solution. Let I be the selected set of r rows and J the selected set of s columns. In a permutation product, the rows in I use r distinct columns. Only $n - s = r - 1$ columns lie outside J , so at least one of those rows must use a column in J . Its entry is zero. Every permutation product therefore vanishes, and their signed sum is zero.

Exercise 5.6 (An integer matrix and its inverse). Let A be an invertible $n \times n$ matrix. If every entry of A and A^{-1} is an integer, prove that $\det A = \pm 1$.

Solution. The permutation formula makes $\det A$ and $\det A^{-1}$ integers. Their product is $\det I_n = 1$. Two integers with product 1 are both 1 or both -1 : their nonzero absolute values are positive integers whose product is 1. Thus $\det A = \pm 1$.

Exercise 5.7 (Cofactors of a symmetric matrix). Prove that the cofactor matrix of a symmetric matrix is symmetric.

Solution. For $A = A^\top$, the ordered submatrices obtained by deleting row i , column j , or row j , column i , satisfy $A_{(ij)}^\top = A_{(ji)}$. Therefore

$$C_{ij} = (-1)^{i+j} \det A_{(ij)} = (-1)^{j+i} \det A_{(ji)} = C_{ji}.$$

For order 1 both minors are the empty matrix, so the same convention applies.

Exercise 5.8 (Adjugate identities*). Let A, B be $n \times n$ matrices. Prove the following.

- (a) $A \operatorname{adj}(A) = \operatorname{adj}(A)A = (\det A)I_n$.
- (b) If A is invertible, then $A^{-1} = \operatorname{adj}(A)/\det A$.
- (c) For $n \geq 2$, $\det \operatorname{adj}(A) = (\det A)^{n-1}$. For $n = 1$, $\det \operatorname{adj}(A) = 1$.
- (d) $\operatorname{adj}(AB) = \operatorname{adj}(B) \operatorname{adj}(A)$, including singular A and B .

Solution.

- (a) If C_{ij} are the cofactors of A , then $(A \operatorname{adj}(A))_{ik} = \sum_j a_{ij} C_{kj}$. For $i = k$ this is row i 's cofactor expansion. For $i \neq k$, replace row k by row i ; the cofactors along row k do not change, while the new determinant is zero because two rows agree. Thus $A \operatorname{adj}(A) = (\det A)I_n$. The column version gives $(\operatorname{adj}(A)A)_{ik} = \sum_j C_{ji} a_{jk}$, equal to $\det A$ if $i = k$ and zero otherwise, by replacing column i with column k .
- (b) Invertibility gives $\det A \neq 0$. Dividing both identities in (a) by $\det A$ shows that $\operatorname{adj}(A)/\det A$ is the two-sided inverse.
- (c) Taking determinants in (a) gives

$$(\det A) \det \operatorname{adj}(A) = (\det A)^n.$$

For invertible A , division proves the claim. If $A \neq 0$ is singular and $\operatorname{adj}(A)$ were invertible, $A \operatorname{adj}(A) = 0$ would imply $A = 0$, a contradiction. Hence $\operatorname{adj}(A)$ is singular. If $A = 0$ and $n \geq 2$, all its order $n - 1$ minors are zero, so $\operatorname{adj}(A) = 0$. Thus both sides of the claimed formula are zero in every singular case with $n \geq 2$. For $n = 1$, the empty-minor convention gives $\operatorname{adj}[a] = [1]$ for every a .

- (d) If both matrices are invertible, (b) and the order of an inverse product give

$$\operatorname{adj}(AB) = \det(AB)(AB)^{-1} = (\det B)B^{-1}(\det A)A^{-1} = \operatorname{adj}(B) \operatorname{adj}(A).$$

For singular matrices, use the following polynomial argument.

- For $k \geq 1$, multiplication and cancellation give

$$t^k - a^k = (t - a) \sum_{r=0}^{k-1} t^{k-1-r} a^r.$$

Consequently, for $p(t) = \sum_{k=0}^d c_k t^k$,

$$p(t) - p(a) = (t - a)q(t), \quad q(t) = \sum_{k=1}^d c_k \sum_{r=0}^{k-1} t^{k-1-r} a^r.$$

- If p has degree $d \geq 1$, then q has degree $d - 1$, since its leading coefficient is $c_d \neq 0$. If $p(a) = 0$, every other root of p is a root of q . Induction on d , starting with a nonzero constant having no roots, proves that a nonzero polynomial of degree d has at most d distinct real roots. In particular, a polynomial vanishing at infinitely many real numbers has every coefficient zero.

- Every determinant of a matrix with polynomial entries is a polynomial, by its finite permutation sum, and the same holds for every adjugate entry. Moreover, $\det(A + tI_n)$ is monic of degree n : the term t^n can only come from choosing all diagonal t 's in the identity permutation.
- Thus each of $A + tI_n$ and $B + tI_n$ is invertible except at finitely many real t . For every other t , the already proved invertible case gives

$$\operatorname{adj}\left((A + tI_n)(B + tI_n)\right) = \operatorname{adj}(B + tI_n) \operatorname{adj}(A + tI_n).$$

Each entry of the difference is a polynomial with infinitely many roots, so it is zero for all t . Evaluation at $t = 0$ proves (d).

Exercise 5.9 (Rank and the adjugate). Let A be $n \times n$, $n \geq 2$, with cofactors C_{ij} .

- (a) Suppose $\operatorname{rank} A = n - 1$. For any nonzero columns x, y satisfying $Ax = 0$, $A^\top y = 0$, show that

$$\operatorname{adj}(A) = cxy^\top$$

for a nonzero scalar c . Show also that $c = C_{ij}/(y_i x_j)$ whenever $y_i \neq 0$ and $x_j \neq 0$.

- (b) If $\operatorname{rank} A \leq n - 2$, prove that $\operatorname{adj}(A) = 0$.

Solution. First connect minors to rank.

- If a matrix has rank $r > 0$, use Theorem 3.3 to extract r independent columns and call the resulting matrix F . It has column rank r , hence row rank r by Theorem 3.7. Extract r independent rows of F . They form a square matrix G with row rank r , hence column rank r . The invertibility criterion gives $\det G \neq 0$. Thus the original matrix has a nonzero order r minor.
- Conversely, a nonsingular order k submatrix forces its k selected columns of the original matrix to be independent: any relation among those columns restricts on the selected rows to a relation among the submatrix columns, whose coefficients must all vanish. Therefore $k \leq r$. This also shows that a rank-zero matrix has no nonzero minor of positive order.

- (a) • Theorems 3.11 and 3.7 give $\dim N(A) = \dim N(A^\top) = 1$. Since $x, y \neq 0$,

$$N(A) = \operatorname{span}(x), \quad N(A^\top) = \operatorname{span}(y).$$

- Since $\det A = 0$, Exercise 5.8(a) gives $A \operatorname{adj}(A) = \operatorname{adj}(A)A = 0$. Every column of $\operatorname{adj}(A)$ lies in $N(A) = \operatorname{span}(x)$, so $\operatorname{adj}(A) = xz^\top$ for some z . Choose $x_k \neq 0$. Row k of $xz^\top A = 0$ gives

$$x_k z^\top A = 0 \implies z^\top A = 0 \implies z \in N(A^\top) = \operatorname{span}(y).$$

Thus $z = cy$ and $\operatorname{adj}(A) = cxy^\top$.

- Some order $n - 1$ minor is nonzero by the argument above. Hence $\operatorname{adj}(A) \neq 0$, so $c \neq 0$. Entry (j, i) gives $C_{ij} = cx_j y_i$; for every $x_j y_i \neq 0$,

$$c = \frac{C_{ij}}{y_i x_j}.$$

- (b) A nonzero order $n - 1$ minor would force $\operatorname{rank} A \geq n - 1$ by the converse just proved. Hence all such minors, and therefore all cofactors, are zero. Their transpose $\operatorname{adj}(A)$ is zero.

Exercise 5.10 (A bordered determinant without invertibility). Let A be $n \times n$, $x, y \in \mathbb{R}^n$, and $c \in \mathbb{R}$.

(a) Prove that

$$\det \begin{pmatrix} A & y \\ x^\top & c \end{pmatrix} = c \det A - x^\top \operatorname{adj}(A)y.$$

(b) If A is invertible, rewrite this as $\det A(c - x^\top A^{-1}y)$. Relate it to the general block identities, for invertible $T : n \times n$, $U : n \times m$, $V : m \times n$, $W : m \times m$,

$$\det \begin{pmatrix} T & U \\ V & W \end{pmatrix} = \det \begin{pmatrix} W & V \\ U & T \end{pmatrix} = \det T \det(W - VT^{-1}U).$$

Solution.

(a) Let $A_{\hat{j}}$ be A with column j deleted, and let $A_{(ij)}$ have row i and column j deleted. Expanding along the last row, then the last column of each remaining minor, gives

$$\begin{aligned} \det \begin{pmatrix} A & y \\ x^\top & c \end{pmatrix} &= c \det A + \sum_{j=1}^n x_j (-1)^{n+1+j} \det[A_{\hat{j}} y], \\ \det[A_{\hat{j}} y] &= \sum_{i=1}^n y_i (-1)^{i+n} \det A_{(ij)}. \end{aligned}$$

The combined sign is $(-1)^{2n+1+i+j} = -(-1)^{i+j}$. The double sum is therefore $-\sum_{i,j} x_j C_{ij} y_i = -x^\top \operatorname{adj}(A)y$, proving (a). This uses no inverse and also holds for $n = 1$.

(b) Substitute $\operatorname{adj}(A) = (\det A)A^{-1}$ in (a). For the general identity, block elimination gives

$$\begin{pmatrix} I_n & 0 \\ -VT^{-1} & I_m \end{pmatrix} \begin{pmatrix} T & U \\ V & W \end{pmatrix} = \begin{pmatrix} T & U \\ 0 & W - VT^{-1}U \end{pmatrix}.$$

The multiplier has determinant 1, giving the last expression. To exchange the two block orders, apply the same permutation to rows and columns. If its permutation matrix is P , the new matrix is PMP^{-1} , whose determinant is $\det M$ by multiplication and $\det P \det P^{-1} = 1$. Setting $T = A, U = y, V = x^\top, W = [c]$ recovers the scalar formula.

Exercise 5.11 (Cramer's rule). Let $A = (a_{ij})$ be invertible of order n , with cofactors C_{ij} . For the solution of $Ax = b$, prove the following.

(a) $x_j = \frac{\sum_{i=1}^n b_i C_{ij}}{\det A}, 1 \leq j \leq n.$

(b) Let $A^{(j)}$ be A with column j replaced by b . Then $x_j = \det A^{(j)} / \det A$.

Solution.

(a) The unique solution is $x = A^{-1}b = (\det A)^{-1} \operatorname{adj}(A)b$. Since entry (j, i) of $\operatorname{adj}(A)$ is C_{ij} , its j th component is the expression in (a).

(b) Deleting row i and column j removes the changed column, so the cofactors in column j of $A^{(j)}$ are still C_{ij} . The column expansion of Theorem 5.7 gives $\det A^{(j)} = \sum_i b_i C_{ij}$. Substitution in (a) proves the ratio.

Exercise 5.12 (Fitting a polynomial*). Let $x_1, \dots, x_{n+1}$ be distinct real numbers and let $y_1, \dots, y_{n+1}$ be prescribed real values. Determine all coefficients of the polynomial

$$p(x) = \sum_{j=0}^n a_j x^j, \quad p(x_i) = y_i \quad (1 \leq i \leq n+1).$$

The polynomial has degree at most n ; its highest coefficients may vanish.

Solution. Use the column Vandermonde matrix

$$V = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ x_1 & x_2 & \cdots & x_{n+1} \\ x_1^2 & x_2^2 & \cdots & x_{n+1}^2 \\ \vdots & \vdots & & \vdots \\ x_1^n & x_2^n & \cdots & x_{n+1}^n \end{pmatrix}, \quad a = (a_0, \dots, a_n)^\top, \quad y = (y_1, \dots, y_{n+1})^\top.$$

The equations are $V^\top a = y$, and Example 5.4 gives

$$\det V = \prod_{1 \leq j < i \leq n+1} (x_i - x_j) \neq 0.$$

Thus there is exactly one coefficient column. For $0 \leq j \leq n$, let V_j be obtained from V by replacing row $j+1$ by $y^\top$. Replacing column $j+1$ of $V^\top$ by y gives $V_j^\top$. Exercise 5.11 and invariance of determinant under transpose now give all coefficients:

$$a_j = \frac{\det V_j}{\prod_{1 \leq k < i \leq n+1} (x_i - x_k)} \quad (0 \leq j \leq n).$$

These coefficients solve $V^\top a = y$, so the resulting polynomial has the required values. Uniqueness of the coefficient column proves uniqueness of the polynomial. For example, zero prescribed values yield all coefficients zero; this is why degree exactly n is not required.

Block elimination and determinant identities

Exercise 5.13 (The two block inverse formulas). Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, where $A : m \times m$, $B : m \times n$, $C : n \times m$, $D : n \times n$. Prove the following formulas under their respective hypotheses.

(a) If A and $E = D - CA^{-1}B$ are invertible, then

$$M^{-1} = \begin{pmatrix} A^{-1} + A^{-1}BE^{-1}CA^{-1} & -A^{-1}BE^{-1} \\ -E^{-1}CA^{-1} & E^{-1} \end{pmatrix}.$$

(b) If D and $F = A - BD^{-1}C$ are invertible, then

$$M^{-1} = \begin{pmatrix} F^{-1} & -F^{-1}BD^{-1} \\ -D^{-1}CF^{-1} & D^{-1} + D^{-1}CF^{-1}BD^{-1} \end{pmatrix}.$$

Solution.

(a) Direct block multiplication gives

$$LM = N, \quad L = \begin{pmatrix} I_m & 0 \\ -CA^{-1} & I_n \end{pmatrix}, \quad N = \begin{pmatrix} A & B \\ 0 & E \end{pmatrix}.$$

The inverse of L is obtained by changing the minus sign to plus. Furthermore,

$$H = \begin{pmatrix} A^{-1} & -A^{-1}BE^{-1} \\ 0 & E^{-1} \end{pmatrix}$$

satisfies $NH = HN = I_{m+n}$: the upper-right blocks are respectively $-BE^{-1} + BE^{-1} = 0$ and $A^{-1}B - A^{-1}B = 0$, and the diagonal blocks are identities. Thus $M^{-1} = HL$. Multiplication gives

$$HL = \begin{pmatrix} A^{-1} + A^{-1}BE^{-1}CA^{-1} & -A^{-1}BE^{-1} \\ -E^{-1}CA^{-1} & E^{-1} \end{pmatrix},$$

as required.

(b) With

$$L = \begin{pmatrix} I_m & -BD^{-1} \\ 0 & I_n \end{pmatrix}, \quad N = \begin{pmatrix} F & 0 \\ C & D \end{pmatrix},$$

we have $LM = N$; again L^{-1} changes the minus sign to plus. Set

$$H = \begin{pmatrix} F^{-1} & 0 \\ -D^{-1}CF^{-1} & D^{-1} \end{pmatrix}.$$

The lower-left blocks of NH and HN are $CF^{-1} - CF^{-1} = 0$ and $-D^{-1}C + D^{-1}C = 0$, respectively; the other blocks give $NH = HN = I_{m+n}$. Consequently

$$M^{-1} = HL = \begin{pmatrix} F^{-1} & -F^{-1}BD^{-1} \\ -D^{-1}CF^{-1} & D^{-1} + D^{-1}CF^{-1}BD^{-1} \end{pmatrix}.$$

Both constructions prove existence of the inverse under exactly the stated hypotheses, as well as identifying its entries.

Exercise 5.14 (Two diagonal zero blocks*). (a) Prove that

$$\det \begin{pmatrix} 0_{m \times n} & I_m \\ I_n & 0_{n \times m} \end{pmatrix} = (-1)^{mn}.$$

(b) Use (a) to prove, for square $B : m \times m$ and $C : n \times n$,

$$\det \begin{pmatrix} 0_{m \times n} & B \\ C & 0_{n \times m} \end{pmatrix} = (-1)^{mn} \det B \det C.$$

(c) Explain why the answer in (b) becomes $(-1)^m \det B \det C$ when $m = n$.

(d) What happens when B and C are not square, but $Z = \begin{pmatrix} 0 & B \\ C & 0 \end{pmatrix}$ is square?

Solution.

(a) Denote the matrix in (a) by Z . Move the last m columns, in their existing order, before the first n columns. Each of those m columns crosses exactly n columns, so there are mn adjacent interchanges. The resulting matrix is I_{m+n} . Since every interchange changes determinant sign, $1 = (-1)^{mn} \det Z$, and hence $\det Z = (-1)^{mn}$.

(b) Factor the matrix as

$$\begin{pmatrix} 0 & B \\ C & 0 \end{pmatrix} = \begin{pmatrix} B & 0 \\ 0 & C \end{pmatrix} \begin{pmatrix} 0 & I_m \\ I_n & 0 \end{pmatrix}.$$

Determinant multiplication, the block-diagonal formula, and (a) give the claim.

- (c) The integer $m^2 - m = m(m - 1)$ is even, since one of two consecutive integers is even. Thus $(-1)^{m^2} = (-1)^m$.
- (d) Write $B : r \times s$, $C : u \times v$, where $r + u = s + v$ and $r \neq s$. If $s > r$, the s columns of B are dependent, so $Bz = 0$ for some nonzero $z \in \mathbb{R}^s$. Then

$$Z \underbrace{\begin{pmatrix} 0_v \\ z \end{pmatrix}}_{\neq 0} = 0,$$

so Z has a nonzero null vector. If $s < r$, the size equality gives $v > u$, so $Cw = 0$ for some nonzero $w \in \mathbb{R}^v$, and $Z \begin{pmatrix} w \\ 0_s \end{pmatrix} = 0$. Thus Z is singular in both cases and its determinant is zero.

Exercise 5.15 (Eliminating a product block). For $B : m \times n$ and any $D : n \times n$, prove that

$$\det \begin{pmatrix} I_m & B \\ B^\top & B^\top B + D \end{pmatrix} = \det D.$$

Solution. The block column operation is the explicit product

$$\begin{pmatrix} I_m & B \\ B^\top & B^\top B + D \end{pmatrix} \begin{pmatrix} I_m & -B \\ 0 & I_n \end{pmatrix} = \begin{pmatrix} I_m & 0 \\ B^\top & D \end{pmatrix}.$$

The multiplier has determinant 1; the resulting lower block-triangular matrix has determinant $\det I_m \det D = \det D$. Taking determinants proves the identity without an invertibility assumption on D .

Exercise 5.16 (One diagonal zero block). (a) If $A : m \times m$ is invertible and $b, c \in \mathbb{R}^m$, prove

$$\det \begin{pmatrix} A & b \\ c^\top & 0 \end{pmatrix} = -\det A (c^\top A^{-1} b).$$

(b) Generalize (a): for invertible $A : m \times m$, $B : m \times n$, $C : n \times m$, prove

$$\det \begin{pmatrix} A & B \\ C & 0_{n \times n} \end{pmatrix} = (-1)^n \det A \det(CA^{-1}B).$$

(c) For invertible $D : n \times n$, $B : m \times n$, $C : n \times m$, prove

$$\det \begin{pmatrix} 0_{m \times m} & B \\ C & D \end{pmatrix} = (-1)^m \det D \det(BD^{-1}C).$$

Solution.

(a) The determinant-one multiplier gives

$$\begin{pmatrix} I_m & 0 \\ -c^\top A^{-1} & 1 \end{pmatrix} \begin{pmatrix} A & b \\ c^\top & 0 \end{pmatrix} = \begin{pmatrix} A & b \\ 0 & -c^\top A^{-1} b \end{pmatrix}.$$

Its determinant is the product $-\det A (c^\top A^{-1} b)$.

(b) Similarly,

$$\begin{pmatrix} I_m & 0 \\ -CA^{-1} & I_n \end{pmatrix} \begin{pmatrix} A & B \\ C & 0 \end{pmatrix} = \begin{pmatrix} A & B \\ 0 & -CA^{-1}B \end{pmatrix}.$$

The right-hand determinant is $\det A \det(-CA^{-1}B)$. Factoring -1 from each of its last block's n rows gives the sign $(-1)^n$.

(c) The appropriate multiplier is now upper triangular:

$$\begin{pmatrix} I_m & -BD^{-1} \\ 0 & I_n \end{pmatrix} \begin{pmatrix} 0 & B \\ C & D \end{pmatrix} = \begin{pmatrix} -BD^{-1}C & 0 \\ C & D \end{pmatrix}.$$

Its determinant is $\det(-BD^{-1}C) \det D$, which gives the claimed $(-1)^m$ factor because $BD^{-1}C$ has order m .

Exercise 5.17 (Scalar pivots and matrix pivots). Let $A : m \times m$, $b, c \in \mathbb{R}^m$, $D : n \times n$, $u, v \in \mathbb{R}^n$, and let α, δ be scalars. Set

$$Z_1 = \begin{pmatrix} A & b \\ c^\top & \delta \end{pmatrix}, \quad Z_2 = \begin{pmatrix} \alpha & u^\top \\ v & D \end{pmatrix}.$$

Prove each formula using only its stated invertibility hypothesis.

- (a) If A is invertible, $\det Z_1 = \det A (\delta - c^\top A^{-1}b)$.
- (b) If $\delta \neq 0$, $\det Z_1 = \delta \det(A - \delta^{-1}bc^\top)$.
- (c) If D is invertible, $\det Z_2 = \det D (\alpha - u^\top D^{-1}v)$.
- (d) If $\alpha \neq 0$, $\det Z_2 = \alpha \det(D - \alpha^{-1}vu^\top)$.

Solution. The following four products keep the assumptions separate:

(a)

$$\begin{pmatrix} I_m & 0 \\ -c^\top A^{-1} & 1 \end{pmatrix} Z_1 = \begin{pmatrix} A & b \\ 0 & \delta - c^\top A^{-1}b \end{pmatrix}.$$

(b)

$$\begin{pmatrix} I_m & -\delta^{-1}b \\ 0 & 1 \end{pmatrix} Z_1 = \begin{pmatrix} A - \delta^{-1}bc^\top & 0 \\ c^\top & \delta \end{pmatrix}.$$

(c)

$$\begin{pmatrix} 1 & -u^\top D^{-1} \\ 0 & I_n \end{pmatrix} Z_2 = \begin{pmatrix} \alpha - u^\top D^{-1}v & 0 \\ v & D \end{pmatrix}.$$

(d)

$$\begin{pmatrix} 1 & 0 \\ -\alpha^{-1}v & I_n \end{pmatrix} Z_2 = \begin{pmatrix} \alpha & u^\top \\ 0 & D - \alpha^{-1}vu^\top \end{pmatrix}.$$

Every left multiplier is block triangular with identity diagonal blocks, so its determinant is 1. Every resulting matrix is block triangular; taking the product of its diagonal-block determinants yields the four formulas.

Exercise 5.18 (Operations on entire block rows). Let $M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$, with $A : m \times m$, $B : m \times n$, $C : n \times m$, $D : n \times n$.

(a) For every $E : m \times m$, prove

$$\det \begin{pmatrix} EA & EB \\ C & D \end{pmatrix} = \det E \det M.$$

(b) For every $E : n \times m$, prove

$$\det \begin{pmatrix} A & B \\ C + EA & D + EB \end{pmatrix} = \det M.$$

Solution.

(a) Multiplication by $\text{diag}(E, I_n)$ gives exactly $\begin{pmatrix} EA & EB \\ C & D \end{pmatrix}$. The multiplier has determinant $\det E$; determinant multiplication proves the formula, even if E is singular.

(b) Directly,

$$\begin{pmatrix} I_m & 0 \\ E & I_n \end{pmatrix} M = \begin{pmatrix} A & B \\ C + EA & D + EB \end{pmatrix}.$$

The multiplier's determinant is 1, so the determinant is unchanged.

Exercise 5.19 (A diagonal block of an inverse). Let

$$Z = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad Z^{-1} = \begin{pmatrix} P & Q \\ R & S \end{pmatrix},$$

where $A, P : m \times m$, $D, S : n \times n$, and the remaining blocks are conformable. Assume only that Z is invertible; A and D may be singular. Prove

$$\det S = \frac{\det A}{\det Z}, \quad \det P = \frac{\det D}{\det Z}.$$

Solution. The four blocks of $ZZ^{-1} = I_{m+n}$ are

$$AP + BR = I_m, \quad AQ + BS = 0, \quad CP + DR = 0, \quad CQ + DS = I_n.$$

Consequently,

$$Z \begin{pmatrix} I_m & Q \\ 0 & S \end{pmatrix} = \begin{pmatrix} A & 0 \\ C & I_n \end{pmatrix}, \quad Z \begin{pmatrix} P & 0 \\ R & I_n \end{pmatrix} = \begin{pmatrix} I_m & B \\ 0 & D \end{pmatrix}.$$

Taking determinants gives $\det Z \det S = \det A$ and $\det Z \det P = \det D$. Division by $\det Z \neq 0$ proves both identities. No inverse of A, D, P , or S was used.

Exercise 5.20 (When two blocks commute*). Let A, B, C, D all have order n . If $AC = CA$, prove

$$\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \det(AD - CB).$$

Include the case where A is singular.

Solution.

- If A is invertible, Theorems 5.10 and 5.5 give

$$\begin{aligned} \det \begin{pmatrix} A & B \\ C & D \end{pmatrix} &= \det A \det(D - CA^{-1}B) \\ &= \det(AD - ACA^{-1}B) = \det(AD - CB), \end{aligned}$$

since $ACA^{-1} = CAA^{-1} = C$.

- For arbitrary A , put $A_t = A + tI_n$. Since $A_tC = AC + tC = CA + tC = CA_t$, it still commutes with C . The polynomial argument proved in Exercise 5.8(d) shows that $\det A_t$ is monic and nonzero except at finitely many real t . For every other t , the invertible case proves

$$\det \begin{pmatrix} A + tI_n & B \\ C & D \end{pmatrix} = \det(AD - CB + tD).$$

Both sides are polynomials in t , because their matrix entries are polynomials. Their difference vanishes at infinitely many real numbers; by the root bound already proved, it is the zero polynomial. Set $t = 0$ to obtain the identity, including singular A .

Exercise 5.21 (A determinant after a matrix update). Let $A : m \times m$ and $D : n \times n$ be invertible, and let $B : m \times n$, $C : n \times m$. Use Sylvester's determinant identity to prove

$$\det(A + BDC) = \det A \det D \det(D^{-1} + CA^{-1}B).$$

Solution. Apply $\det(I_m + UV) = \det(I_n + VU)$ from Theorem 5.11 to $U = A^{-1}B : m \times n$, $V = DC : n \times m$. Then

$$\begin{aligned} \det(A + BDC) &= \det A \det(I_m + A^{-1}BDC) \\ &= \det A \det(I_n + DCA^{-1}B) \\ &= \det A \det(D(D^{-1} + CA^{-1}B)) \\ &= \det A \det D \det(D^{-1} + CA^{-1}B). \end{aligned}$$

The factors inside the last determinant retain their original order.

Exercise 5.22 (A block difference of squares). For any two $n \times n$ matrices A, B , prove

$$\det \begin{pmatrix} A & B \\ B & A \end{pmatrix} = \det(A + B) \det(A - B).$$

Solution. Let Z denote the displayed block matrix. Direct multiplication gives

$$\begin{pmatrix} I_n & I_n \\ 0 & I_n \end{pmatrix} Z = \begin{pmatrix} A + B & A + B \\ B & A \end{pmatrix}$$

and the factorization

$$\begin{pmatrix} A + B & A + B \\ B & A \end{pmatrix} = \begin{pmatrix} A + B & 0 \\ 0 & I_n \end{pmatrix} \begin{pmatrix} I_n & 0 \\ B & I_n \end{pmatrix} \begin{pmatrix} I_n & I_n \\ 0 & A - B \end{pmatrix}.$$

The first multiplier has determinant 1. The three factors on the right have determinants $\det(A + B)$, 1, and $\det(A - B)$, respectively. Taking determinants proves the claim. None of these products requires $AB = BA$, or invertibility of $A + B$ or $A - B$.

Exercise 5.23 (Three groups of coordinates). Let $A : m \times m$, $D : n \times n$, $E : p \times p$ be symmetric, and let $B : m \times n$, $C : m \times p$. If D, E are invertible, prove

$$\det \begin{pmatrix} A & B & C \\ B^\top & D & 0 \\ C^\top & 0 & E \end{pmatrix} = \det D \det E \det(A - BD^{-1}B^\top - CE^{-1}C^\top).$$

Solution. Combine the last two coordinate groups:

$$F = [B \ C], \quad H = \begin{pmatrix} D & 0 \\ 0 & E \end{pmatrix}, \quad H^{-1} = \begin{pmatrix} D^{-1} & 0 \\ 0 & E^{-1} \end{pmatrix}.$$

The formula for H^{-1} follows by multiplying in both orders. The original matrix is $\begin{pmatrix} A & F \\ F^\top & H \end{pmatrix}$. Theorem 5.10, using the invertible lower-right block, yields

$$\det H \det(A - FH^{-1}F^\top).$$

Here $\det H = \det D \det E$, and block multiplication gives

$$FH^{-1}F^\top = [BD^{-1} \ CE^{-1}] \begin{pmatrix} B^\top \\ C^\top \end{pmatrix} = BD^{-1}B^\top + CE^{-1}C^\top.$$

Substitution proves the required determinant.

Exercise 5.24 (A scalar at the opposite corner). Let A be invertible of order m , let $a \in \mathbb{R}^m$, and let $\alpha \neq 0$. Prove

$$\det \begin{pmatrix} 0_{m \times 1} & A \\ \alpha & a^\top \end{pmatrix} = (-1)^m \alpha \det A.$$

Solution. Column 1 has only the bottom entry α nonzero. Deleting its row $m+1$ and column 1 leaves exactly A , so expansion gives

$$\det \begin{pmatrix} 0 & A \\ \alpha & a^\top \end{pmatrix} = \alpha (-1)^{(m+1)+1} \det A = (-1)^m \alpha \det A.$$

Exercise 5.25 (A block identity without a pivot). For arbitrary $n \times n$ matrices A, B , prove

$$\det \begin{pmatrix} A+B & A \\ A & A \end{pmatrix} = \det A \det B.$$

Solution. Subtract the second block row from the first by multiplying:

$$\begin{pmatrix} I_n & -I_n \\ 0 & I_n \end{pmatrix} \begin{pmatrix} A+B & A \\ A & A \end{pmatrix} = \begin{pmatrix} B & 0 \\ A & A \end{pmatrix}.$$

The multiplier has determinant 1. The lower block-triangular matrix has determinant $\det B \det A$, which equals $\det A \det B$ because these two factors are real scalars. This proves the formula also when A or B is singular.